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Inventor(s)	Alghanmi; Abdulmalik Abdullah et al.

Advanced drone port station for enhanced maintenance, monitoring, and surveillance operations

Abstract

A drone system includes a headquarter station and one or more drone port stations. Each drone port station includes a platform configured to receive a drone, a radar system configured to detect and identify the drone, and one or more locker clips disposed on the platform, wherein the one or more locker clips are configured to lock onto one or more legs extending from the drone. Each drone port station also includes an internal computer configured to control operations of the drone port station, an internal camera operatively connected to the internal computer, where the internal camera is configured to detect a charging port or a fuel tank of the drone, a glass panel placed on the platform, where the glass panel is configured to protect the internal camera, and an antenna extending from the platform and configured to provide a communications pathway from the drone port station to the headquarter station.

Inventors: **Alghanmi; Abdulmalik Abdullah (Al Ghazawat, SA), Al-Qasim; Abdulaziz S. (Dammam, SA)**

Applicant: **SAUDI ARABIAN OIL COMPANY** (Dhahran, SA); **KING FAHD UNIVERSITY OF PETROLEUM & MINERALS** (Dhahran, SA)

Family ID: **90105459**

Assignee: **SAUDI ARABIAN OIL COMPANY (Dhahran, SA); KING FAHD UNIVERSITY OF PETROLEUM & MINERALS (Dhahran, SA)**

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Primary Examiner: Green; Richard

Attorney, Agent or Firm: Osha Bergman Watanabe & Burton LLP

Background/Summary

BACKGROUND
(1) Drones are commonly used to automate tasks related to transposition, delivery, maintenance, monitoring, and surveillance. Drones may be pre-programmed or controlled autonomously from a centralized headquarter (HQ) station. In many situations, drones are powered by an on-board battery. Such drones are typically suited to environments which are relatively compact, such as a city or other urban populated area.
SUMMARY
(2) This summary is provided to introduce a selection of concepts that are further described below in the detailed description. This summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used as an aid in limiting the scope of the claimed subject matter.
(3) In one aspect, embodiments disclosed herein relate to a drone system which may include a headquarter station and one or more drone port stations. Each drone port station may include a platform configured to receive a drone, a radar system configured to detect and identify the drone, and one or more locker clips disposed on the platform, wherein the one or more locker clips are configured to lock onto one or more legs extending from the drone. Each drone port station may also include an internal computer configured to control operations of the drone port station, an internal camera operatively connected to the internal computer, where the internal camera is configured to detect a charging port or a fuel tank of the drone, a glass panel placed on the platform, where the glass panel is configured to protect the internal camera, and an antenna extending from the platform and configured to provide a communications pathway from the drone port station to the headquarter station.
(4) In another aspect, embodiments disclosed herein relate to a drone port station. The drone port station may include a platform configured to receive a drone, a radar system configured to detect and identify the drone, and one or more locker clips disposed on the platform, wherein the one or more locker clips are configured to lock onto one or more legs extending from the drone. The drone port station may also include an internal computer configured to control operations of the drone port station, an internal camera operatively connected to the internal computer, where the internal camera is configured to detect a charging port or a fuel tank of the drone, and a fueling conduit configured to connect to a fueling port of a drone. The drone port station may further include a glass panel placed on the platform, where the glass panel is configured to protect the internal camera, and an antenna extending from the platform.
(5) In yet another aspect, embodiments disclosed herein relate to a method of charging a drone at a drone port station. The method may include detecting the drone using a radar system disposed on the drone port station, guiding, using the radar system, the drone to a platform of the drone port

station, and locking the drone onto the platform using one or more locker clips. The method may also include detecting a fueling port of the drone using an internal camera disposed on the drone port station, and connecting a fueling conduit to the fueling port of the drone, where the fueling port is a charging port or a fuel tank, and where the fueling conduit is a movable charger or a fuel pipe.
(6) Other aspects and advantages of the claimed subject matter will be apparent from the following description and the appended claims.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) Specific embodiments of the disclosed technology will now be described in detail with reference to the accompanying figures. Like elements in the various figures are denoted by like reference numerals for consistency. The size and relative positions of elements in the drawings are not necessarily drawn to scale. For example, the shapes of various elements and angles are not necessarily drawn to scale, and some of these elements may be arbitrarily enlarged and positioned to improve drawing legibility. Further, the particular shapes of the elements as drawn are not necessarily intended to convey any information regarding the actual shape of the particular elements and have been solely selected for ease of recognition in the drawing.
(2) FIG. **1** shows a drone system in accordance with one or more embodiments.
(3) FIGS. **2**A-**2**B show a drone port station in accordance with one or more embodiments.
(4) FIG. **3** shows a self-cleaning solar panel in accordance with one or more embodiments.
(5) FIG. **4** shows a drone port station in accordance with one or more embodiments.
(6) FIG. **5** shows a drone port station in accordance with one or more embodiments.
(7) FIG. **6** shows a flowchart of a method in accordance with one or more embodiments.
DETAILED DESCRIPTION
(8) In the following detailed description of embodiments of the disclosure, numerous specific details are set forth in order to provide a more thorough understanding of the disclosure. However, it will be apparent to one of ordinary skill in the art that the disclosure may be practiced without these specific details. In other instances, well-known features have not been described in detail to avoid unnecessarily complicating the description.
(9) Throughout the application, ordinal numbers (e.g., first, second, third, etc.) may be used as an adjective for an element (i.e., any noun in the application). The use of ordinal numbers is not to imply or create any particular ordering of the elements nor to limit any element to being only a single element unless expressly disclosed, such as using the terms “before”, “after”, “single”, and other such terminology. Rather, the use of ordinal numbers is to distinguish between the elements. By way of an example, a first element is distinct from a second element, and the first element may encompass more than one element and succeed (or precede) the second element in an ordering of elements.
(10) In the following description of FIGS. **1-6**, any component described with regard to a figure, in various embodiments disclosed herein, may be equivalent to one or more like-named components described with regard to any other figure. For brevity, descriptions of these components may not be repeated for each figure. Thus, each and every embodiment of the components of each figure is incorporated by reference and assumed to be optionally present within every other figure having one or more like-named components. Additionally, in accordance with various embodiments disclosed herein, any description of the components of a figure is to be interpreted as an optional embodiment which may be implemented in addition to, in conjunction with, or in place of the embodiments described with regard to a corresponding like-named component in any other figure.
(11) In one aspect, embodiments disclosed herein relate to a drone port station located at a remote

location and configured to receive and charge a drone. In another aspect, embodiments disclosed herein relate to a drone system, where a drone may travel between a headquarter station and a drone port station. In yet another aspect, embodiments disclosed herein relate to a method of charging a drone at a drone port station.

(12) FIG. **1** shows a drone system **100** in accordance with one or more embodiments. The drone system **100** may include a headquarter (HQ) station **102** centrally located at an easily accessible location. For example, the HQ station **102** may be located at an oil and gas processing plant or a drilling site. Alternatively, the HQ station **102** may be located at corporate buildings. The HQ station **102** may be configured to dispatch drones and to monitor the drones during their travelling operations. In one or more embodiments, the HQ station **102** may be configured to fully control drones remotely in situations where manual control is required. The HQ station **102** may be further configured to process images and live videos transmitted by the drones.

(13) The drone system **100** may also include one or more drone port stations **104**. Each drone port station **104** may be located at a distance **105** from the HQ station **102**. Further, each drone port station **104** may include an antenna **106**. A communications pathway **108** may extend from the HQ station **102** to the antenna **106** disposed on each drone port station **104**.

(14) The type of drone port station **104** may differ depending on the desired placement location. A remote area station (RAS) drone port station may be installed in extremely remote locations. An electric transmission lined linked station (ETLLS) drone port station may be built on or near electric transmission lines towers. A fuel station (FS) drone port station may be built near gas stations.

(15) In one or more embodiments, a drone may leave the HQ station **102** and may begin its daily duty. A drone's daily duty may be, for example, monitoring a certain sector of pipelines. When the drone's battery depletes below a certain percentage of its full capacity, the drone may search for the closest drone port station **104** and may communicate with the drone port station **104** via the antenna **106**. The depletion percentage may be selected based on the implementation of the system, the type of drones used, the capacity of the drone port stations **104**, and the distance between adjacent drone port stations **104**. For example, in one or more embodiments, the drone may search for the closest drone port station **104** when the drone's battery depletes below 30% of full capacity. In one or more embodiments, the drone port station **104** may transmit information regarding surrounding conditions, such as weather conditions, to the drone so that the drone may determine if it is safe to fly to the drone port station **104**. If yes, the drone will fly towards the drone port station **104**. In one or more embodiments, the HQ station **102** may be configured to fully control the drone port stations **104** and the drones remotely in situations where manual control is required.

(16) FIGS. **2**A-**2**B show a drone port station in accordance with one or more embodiments. More specifically, FIGS. **2**A-**2**B show a RAS drone port station **200** in accordance with one or more embodiments. The RAS drone port station **200** may include a platform **202** configured to receive a drone. One or more locker clips **204** may be disposed on the platform **202**. The one or more locker clips **204** may be used to lock onto one or more legs which may extend from the drone after the drone has landed on the platform **202**. The locker clips **204** may prevent the drone from slipping and falling from the platform **202**. The locker clips **204** may unlock following completion of the charging procedure, releasing the drone from the platform **202**.

(17) A radar system **208** may extend from the platform **202**. The radar system **208** may detect and identify the drone. Once the radar system **208** has detected the drone, pertinent information regarding the drone may be transmitted to an internal computer **210** disposed on the RAS drone port station **200**. The internal computer **210** may communicate with the HQ stations **102** via an antenna **106** extending from the platform **202**. The internal computer **210** may also control operations of the RAS drone port station **200**.

(18) In one or more embodiments, one or more small turbines **212** may be installed on the platform **202**. When the drone is in the process of landing on the platform **202**, the one or more small

turbines **212** may capture and utilize the air propelled by the landing of the drone to generate energy, which may be stored in a battery **214**.

(19) An internal camera **216** may be operatively connected to the internal computer **210**. A glass panel may be placed on the platform **202** to protect the internal camera **216**. In one or more embodiments, the internal camera **216** may detect the drone's charging port. Once the charging port has been detected, a fueling conduit, such as the movable charger **218** shown in FIG. **2**B, may connect itself to the charging port. Once the drone is fully charged, the movable charger **218** may disconnect itself and the locker clips **204** may unlock, allowing the drone to disengage from the platform **202**.

(20) Referring back to FIG. **2**A, the RAS drone port station **200** may also include a number of ancillary systems. For example, in some embodiments, the RAS drone port station **200** may include a safety lights system **220**. In other embodiments, the RAS drone port station **200** may include a bird repellant system **222**.

(21) The RAS drone port station **200** may be powered by multiple renewable energy sources. The small turbines **212** and one or more main turbines **224** may generate kinetic energy from wind produced by the landing and the taking off of the drone from the platform **202**. One or more solar panels **226** may be installed on the RAS drone port station **200** and may generate solar energy. In one or more embodiments, the solar panels **226** may be fixed, floating, or solar tracker panels. Further, in other embodiments, the solar panels **226** may be self-cleaning solar panels. The RAS drone port station **200** may also be connected to a geothermal plant **228**, such that geothermal energy generated in the geothermal plant **228** may be transmitted to the RAS drone port station **200**.

(22) Since the RAS drone port station **200** may be located in remote areas, such as a desert, for example, an emergency button **230** may be installed on the RAS drone port station **200**. The emergency button **230** may be pressed by a person or operator who has become lost. Pushing the emergency button **230** may create an emergency report, which can be transmitted to the HQ station **102** to alert staff. Staff at the HQ station **102** may then verify the validity of the emergency report using an external camera **232** installed on the RAS drone port station **200**. In one or more embodiments, verifying the validity of the emergency report may include using the external camera **232** to photograph and identify the lost person.

(23) Turning now to FIG. **3**, FIG. **3** shows a self-cleaning solar panel **300** in accordance with one or more embodiments. The solar panels **226**, according to one or more embodiments, may be self-cleaning solar panels **300**. The self-cleaning solar panel **300** may include a surface **302**, where the surface **302** has a first edge **304** and a second edge **306**, where a top edge **308** and a bottom edge **310** extend between the first edge **304** and the second edge **306**. A center **312** may be located equidistant between the first edge **304** and the second edge **306**.

(24) The self-cleaning solar panel **300** may include a mist generator **314** installed along the top edge **308**. The mist generator **314** may collect fluid from the air (i.e., during humid periods) into a mini water tank which may be installed on a rear side of the surface **302**. The mist generator **314** may be configured to dispense a mist of fluid droplets, such that they may flow from the top edge **308** to the bottom edge **310**. In one or more embodiments, the mist generator **314** may be configured to clean dust from the surface **302**.

(25) In one or more embodiments, a steam generator **316** may be installed along a first edge **304** of the surface **302**. In other embodiments, the steam generator **316** may be installed along the second edge **306**, the top edge **308**, or the bottom edge **310**. The steam generator **316** may be configured to remove sticky substances from the surface **302**.

(26) Two micro wiper blades **318** are installed in the center **312** of the surface **302**, such that each micro wiper blade **318** may clean half of the surface **302**. Specifically, each of the micro wiper blade **318** may move from the center **312** of the surface **302** to the first edge **304** and the second edge **306** in a first direction and a second direction, respectively. The first direction and the second

direction may be opposite to one another. Use of the two micro wiper blades **318** may allow for cleaning of the surface **302** of the self-cleaning solar panel **300**, increasing its efficiency.

(27) An air blower **320** may installed along the first edge **304**, the second edge **306**, the top edge **308**, or the bottom edge **310**. The air blower **320** may be configured to blow dust off the surface **302**. A micro-tank **322** may be installed along the rear side of the surface **302** and may contain a volume of washing chemicals. Washing chemicals may be dispensed from the micro-tank **322** onto the surface **302**, such that a combination of the washing chemicals, the fluid droplets dispensed by the mist generator **314**, and the micro wiper blades **318** may clean the surface **302**. In one or more embodiments, the washing chemicals may be soap or other detergents. A drain system **324** may be disposed along the bottom edge **310** to collect and remove any excess fluids from the surface **302**.

(28) One or more sensors **326** may be installed on the surface **302**. The sensors **326** may be configured to provide an alert of a need for cleaning and to provide a solar panel efficiency percentage. A cut off may be applied by an operator to trigger a time and method for cleaning. In one or more embodiments, the self-cleaning solar panel **300** may be triggered via a controller **328**. The controller **328** may be, for example, a built-in automated system, a remote-control unit, or a manual trigger.

(29) In one or more embodiments, the RAS drone port station **200** may be located in an area with plenty of sun but extremely low temperatures. In such embodiments, the self-cleaning solar panel **300** may be equipped with a nano-heating system to melt ice which may form on the surface **302**. In some embodiments, solar-magnifying tools or other tools may also be used to melt ice build-up on the surface **302**. Melting of ice may allow for an increase in solar panel response and efficiency.

(30) Turning now to FIG. **4**, FIG. **4** shows a drone port station in accordance with one or more embodiments. More specifically, FIG. **4** shows an electric transmission lines linked station (ETLLS) drone port station **400** in accordance with one or more embodiments. ETLLS drone port station **400** may be built on or near one or more electric transmission lines towers **402**. The ETLLS drone port station **400** may be powered either from the electric field surrounding the energized wires or from the magnetic field produced by current flowing through the wires.

(31) Similar to the RAS drone port station **200**, the ETLLS drone port station **400** may have a platform **202**, with an antenna **106** and a radar system **208** extending from the platform **202**. Using the radar system **208**, the ETLLS drone port station **400** may detect and guide the drone onto the platform **202**. Locker clips **204** may secure the drone to the platform **202** and a movable charger **218** may be used to connect to the drone's charging port and perform a charging operation. The internal camera **216** may be used to assist in detection of the drone's charging port. Once the drone is fully charged, the locker clips **204** may be unlocked, and the drone may take off from the platform.

(32) FIG. **5** shows a drone port station in accordance with one or more embodiments. More specifically, FIG. **5** shows a fuel station (FS) drone port station **500**. FS drone port stations **500** may be built near gas stations **502**. The FS drone port station **500** may be similar in structure and components to the RAS drone port station **200** and the ETLLS drone port station **400**. In one or more embodiments, the fueling conduit installed on the FS drone port station **500** may be a movable fuel pipe **504**, which may be configured to connect to a drone's fuel tank.

(33) In one or more embodiments, the drone system **100** may be configured to respond to one or more emergency situations. One possible emergency situation may be a renewable energy source failure on a RAS drone port station **200**. In such embodiments, a specialized drone may be deployed from the HQ station **102**. The specialized drone may be configured to transport an emergency battery. Once the specialized drone has been dispatched from the HQ station **102**, the specialized drone may locate and fly to the nearest ETLLS drone port station **400** or FS drone port station **500**. Once at the closest ETLLS drone port station **400** or FS drone port station **500**, the specialized drone may collect an emergency battery, which has been fully charged at the station. The specialized drone may then fly to the RAS drone port station **200** which has suffered from an

energy failure. The specialized drone may be configured to land on the RAS drone port station **200** and to plug in the emergency battery before returning to the HQ station **102**.
(34) Another emergency situation may occur when unstable weather conditions prevent a drone from landing at a desired drone port station **104**. In such a situation, the drone may attempt to locate another nearby drone port station **104**. If the drone is not able to locate another nearby drone port station **104**, the drone may notify the HQ station **102** and may locate the nearest safe area. In one or more embodiments, a safe area may refer to a place of shelter at which the drone may wait out inclement weather conditions. If the drone lands at a safe area, all non-vital systems may be switched off to conserve power.
(35) It may be possible for the drone to sense a danger while sheltering at the safe area. In such cases, the drone may take off and locate another safe area. Otherwise, the drone may wait at the original safe area until the weather conditions improve and it is safe to fly. The drone may then take off and continue on to its originally selected drone port station **104**.
(36) FIG. **6** depicts a flowchart in accordance with one or more embodiments. More specifically, FIG. **6** depicts a flowchart **600** of a method of charging a drone at a drone port station in accordance with one or more embodiments. Further, one or more blocks in FIG. **6** may be performed by one or more components as described in FIGS. **1-5**. While the various blocks in FIG. **6** are presented and described sequentially, one of ordinary skill in the art will appreciate that some or all of the blocks may be executed in different orders, may be combined, may be omitted, and some or all of the blocks may be executed in parallel. Furthermore, the blocks may be performed actively or passively.
(37) Initially, a drone port station **104** may detect a drone using a radar system **208**, S**602**. In one or more embodiments, the radar system **208** may extend from a platform **202** of the drone port station **104**. The drone port station **104** may be a RAS drone port station **200**, a ETLLS drone port station **400**, or a FS drone port station **500**.
(38) Using the radar system **208**, the drone port station **104** may guide the drone onto the platform **202**, S**604**. The drone may then be locked onto the platform **202** with one or more locker clips **204**, S**606**. The drone's fueling port may be detected using an internal camera **216** disposed on the drone port station **104**, S**608**. In one or more embodiments, the fueling port may be a charging port or a fuel tank. Further, a fueling conduit may be autonomously connected to the fueling port, S**610**. The fueling conduit may be, for example, a movable charger **218** or a movable fuel pipe **504**.
(39) The drone port station **104** may detect when the drone is fully charged and may then disconnect the fueling conduit. The locker clips **204** may be unlocked and the drone may take off from the platform **202**.
(40) In one or more embodiments, when the drone port station **104** is a RAS drone port station **200**, the fueling conduit may be powered by one or more self-cleaning solar panels **300**. Further, an autonomous cleaning operation on the one or more self-cleaning solar panels **300** to remove an accumulation of debris.
(41) Embodiments of the present disclosure may provide at least one of the following advantages. The drone port stations described herein are self-operating stations which can be positioned in populated or remote areas and are designed to charge or fuel drones mid-way through their predetermined flight path. Use of a drone port station therefore allows drones to travel much further than previously allowed without needed to return to the HQ station.
(42) Use of RAS drone port stations allows the stations to be sustainable in remote areas since these stations are fully dependent on renewable energy sources. This maximizes efficiency and reliability of maintenance, transportation, monitoring, and surveillance operations. Additionally, all types of drone port stations form a self-stabilized system in that they are able to support themselves during emergency conditions.
(43) Each type of drone port station is above to identify, communicate with, and guide drones without human intervention. Each drone port station can also communicate with the HQ station to

provide periodic status reports or to report any unusual occurrences. Further drone port stations can be equipped with a variety of ancillary systems to further expand functionality. For example, use of an internal camera to detect a charging port or fuel tank of a drone allows for autonomous connection of a fueling conduit to the drone.
(44) Although only a few example embodiments have been described in detail above, those skilled in the art will readily appreciate that many modifications are possible in the example embodiments without materially departing from this invention. Accordingly, all such modifications are intended to be included within the scope of this disclosure as defined in the following claims. In the claims, means-plus-function clauses are intended to cover the structures described herein as performing the recited function and not only structural equivalents, but also equivalent structures. Thus, although a nail and a screw may not be structural equivalents in that a nail employs a cylindrical surface to secure wooden parts together, whereas a screw employs a helical surface, in the environment of fastening wooden parts, a nail and a screw may be equivalent structures. It is the express intention of the applicant not to invoke 35 U.S.C. § 112(f) for any limitations of any of the claims herein, except for those in which the claim expressly uses the words ‘means for’ together with an associated function.

Claims

1. A drone port station comprising: a platform configured to receive a drone; a radar system configured to detect and identify the drone; one or more locker clips disposed on the platform, wherein the one or more locker clips are configured to lock onto one or more legs extending from the drone; an internal computer configured to control operations of the drone port station; an internal camera operatively connected to the internal computer, wherein the internal camera is configured to detect a charging port or a fuel tank of the drone; a fueling conduit configured to connect to a fueling port of the drone; a glass panel placed on the platform, wherein the glass panel is configured to protect the internal camera; an antenna extending from the platform, wherein the drone port station is a remote area station (RAS) powered by a renewable energy source; and a solar panel configured to charge an internal battery, wherein the solar panel comprises: a mist generator configured to clean dust from a surface of the solar panel by dispensing a mist from a top edge of the solar panel; a steam generator installed along a first edge of the solar panel, wherein the steam generator is configured to remove one or more sticky substances from the surface; two micro wiper blades installed in a center of the surface, where a first of the micro wiper blades moves in a first direction, wherein a second of the micro wiper blades moves in a second direction, and wherein the first direction and the second direction are opposite to one another; an air blower installed along a second edge of the solar panel, wherein the air blower is configured to blow dust off the surface; one or more sensors configured to provide an alert of a need for cleaning and to provide a solar panel efficiency percentage; and a micro-tank installed along a rear side of the surface, wherein the micro-tank is filled with a volume of washing chemicals.
2. The drone port station of claim 1, wherein the drone port station is configured to communicate with a headquarter station via the antenna.
3. The drone port station of claim 1 further comprising a bird repellent system comprising a metallic spike integrated into the drone port station.
4. The drone port station of claim 1 further comprising a safety lights system integrated into the drone port station.
5. The drone port station of claim 1 further comprising: an emergency button configured to create an emergency report and to alert a headquarter station when pressed by an operator; and an external camera configured to photograph the operator in order to verify a validity of the emergency report.
6. The drone port station of claim 1, wherein the solar panel is triggered by a controller selected from a group consisting of a built-in automated controller, a remote control unit, and a manual

trigger.
7. The drone port station of claim 1, wherein the solar panel further comprises a nano-heating system configured to melt ice formed on the surface in a cold environment.
8. A drone system comprising: a headquarter station; and one or more drone port stations, each drone port station comprising: a platform configured to receive a drone; a radar system configured to detect and identify the drone; one or more locker clips disposed on the platform, wherein the one or more locker clips are configured to lock onto one or more legs extending from the drone; an internal computer configured to control operations of the drone port station; an internal camera operatively connected to the internal computer, wherein the internal camera is configured to detect a charging port or a fuel tank of the drone; a glass panel placed on the platform, wherein the glass panel is configured to protect the internal camera; an antenna extending from the platform and configured to provide a communications pathway from the drone port station to the headquarter station, wherein the drone port station is a remote area station (RAS) powered by a renewable energy source; and a solar panel configured to charge an internal battery, wherein the solar panel comprises: a mist generator configured to clean dust from a surface of the solar panel by dispensing a mist from a top edge of the solar panel; a steam generator installed along a first edge of the solar panel, wherein the steam generator is configured to remove one or more sticky substances from the surface; two micro wiper blades installed in a center of the surface, where a first of the micro wiper blades moves in a first direction, wherein a second of the micro wiper blades moves in a second direction, and wherein the first direction and the second direction are opposite to one another; an air blower installed along a second edge of the solar panel, wherein the air blower is configured to blow dust off the surface; one or more sensors configured to provide an alert of a need for cleaning and to provide a solar panel efficiency percentage; and a micro-tank installed along a rear side of the surface, wherein the micro-tank is filled with a volume of washing chemicals.
9. The drone system of claim 8, wherein each of the one or more drone port stations further comprises: an emergency button configured to create an emergency report and to alert a headquarter station when pressed by an operator; and an external camera configured to photograph the operator in order to verify a validity of the emergency report.
10. The drone system of claim 8, wherein each of the one or more drone port stations further comprises a bird repellent system and a safety lights system integrated into the drone port station, the bird repellent system comprising a metallic spike.
11. A method of charging a drone at a drone port station comprising: detecting the drone using a radar system disposed on the drone port station; guiding, using the radar system, the drone to a platform of the drone port station; locking the drone onto the platform using one or more locker clips; detecting a fueling port of the drone using an internal camera disposed on the drone port station; connecting a fueling conduit to the fueling port of the drone, wherein the fueling port is a charging port or a fuel tank, and wherein the fueling conduit is a movable charger or a fuel pipe; providing, via an antenna extending from the platform, a communications pathway from the drone port station to a headquarter station, wherein the drone port station is a remote area station (RAS) powered by a renewable energy source; charging an internal battery, via a solar panel disposed on the drone port station; dispensing a mist from a top edge of the solar panel, via a mist generator, to clean dust from a surface of the solar panel; removing one or more sticky substances from the surface, via a steam generator installed along a first edge of the solar panel; installing two micro wiper blades in a center of the surface, wherein a first of the micro wiper blades moves in a first direction, wherein a second of the micro wiper blades moves in a second direction, and wherein the first direction and the second direction are opposite to one another; blowing dust off the surface, via an air blower installed along a second edge of the solar panel; alerting a need for cleaning and providing a solar panel efficiency percentage, via one or more sensors; and filling a micro-tank installed along a rear side of the surface with a volume of washing chemicals.
12. The method of claim 11 further comprising: detecting the drone is fully charged; disconnecting

the fueling conduit; unlocking the one or more locker clips; and releasing the drone from the platform.